Fiscal Year:	FY 2012	**Task Last Updated:**	FY 04/03/2012
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Project Title:	Microgravity Fluid Separation: Dynaswirl® Bubble Vortex Separator		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20794	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	08/25/2011	**End Date:**	08/24/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wu, Xiongjun (Dynaflow, Inc.) Loraine, Greg Ph.D. (Dynaflow, Inc.)		
Grant/Contract No.:	NNX11AO76A		
Performance Goal No.:			
Performance Goal Text:			

The limits on the amount of liquids and gases carried to space make it imperative to recycle and reuse. On earth, bubbles are relatively easily separable by buoyancy. In microgravity, other external forces such as a centrifugal force must be utilized to separate bubbles from liquids. Two categories of centrifugal force separators exist: one uses active rotation by mechanically spinning the tank. This is very efficient but requires shaft, bearings, and motor. In the second, the tank is fixed and the rotation is induced by eccentric injection of the mixture (Free Vortex Separator or FVS). This FVS has been proven very efficient for mid-range void fraction (50 -80% gas) with an efficiency approaching 100%. However, this FVS is not as efficient for lower void fraction, and there is a need of a separator for low void fractions.
We have developed a phase separator at DYNAFLOW based on a DYNASWIRL® swirl flow nozzle. It consists of two concentric cylinders, the fluid is introduced in the space between the two cylinders, and the swirling flow inside the inner

Task Description:	cylinder is generated with the help of tangential injection slots. With this configuration one can make the vortex circulation intense enough to generate cavitation with a very stable vaporous/gaseous core even at low flow rates. Operation with the high swirl near or slightly below cavitation inception in the vortex core guaranties removal of air even from mixtures with very low void fractions. This is achieved by forcing the bubbles to expand and collect in the vortex core. These are then removed through two two central exit orifices at each end of the cylinder. This configuration was tested successfully during a first reduced-gravity flight and we are planning additional testing this month. We are further developing and experimentally testing this configuration to characterize its gas-liquid separation performance and to improve its efficiency. We are also conducting numerical simulations to help optimize the design and ultimately build a compact system for space validation on the International Space Station (ISS) using the existing Fluid Integrated Rack facility.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As a passive separator device, the DYNASWIRL® phase separator has the advantage of simplicity and dependability. It requires no moving mechanical parts and low power. Compared to other passive separator devices, which usually are not very efficient at low void fraction, the DYNASWIRL® is able to force low void fraction gas out of the liquid into the central core of the vortex by combining the effects of swirl, cavitation, and rectified gas diffusion. In addition to being an effective phase separator for space applications, it will also find wide applications in earth applications where fast separation of low void fraction gas is of concern.
Task Progress:	During this reporting period, we have continued our efforts to improve the performance of the DYNASWIRL® phase separator and expanded its operation modes to enable continuous feed of mixture into the phase separator. We have built a test rig with the improved design that is ready for the upcoming reduced gravity flight test at the end of April. We also continued our efforts to develop a reliable in-line void fraction measurement device based on conductivity measurement. A. Improved Design for Efficient Gas and Liquid Extraction In the prototype that was successfully tested in a reduced gravity test in 2009, an orifice of fixed diameter was used for gas extraction. However, the core size varies with the operation condition depending on the void fraction and flow rate of the gas-liquid mixture, the higher the void fraction and flow rate, the larger the core size and vice versa. The original design with a fixed orifice size thus puts a limit on the flux of gas being extracted from the core and reduces the overall efficiency of gas extraction especially for high void fraction mixtures. The new orifice design utilizing a hydrophobic membrane is implemented to achieve a gas extraction orifice with variable orifice size dictated by the gaseous core size. The hydrophobic membrane allows only gaseous content to pass through it while blocks the liquid content, this unique feature makes this hydrophobic membrane ideal candidate for the gas extraction orifice of variable orifice size: an enlarged gas extraction orifice that is larger than the largest gaseous core even at higher void fraction, the orifice is covered by the hydrophobic membrane, on one end of the membrane, it connects with the gaseous core of the DYNASWIRL® chamber, on the other end of the membrane, it connects with a small vacuum cell. Due to the pressure difference between the gaseous core and the vacuum cell, the gaseous content collected in the gaseous core will cross the membrane and be extracted, however the liquid in the DYNASWIRL® chamber is prevented from entering the vacuum cell. With such a design, smooth gas extraction process is achieved. In the previous prototype design, the liquid exit was concentric with the gaseous core in the opposite end of the DYNASWIRL® chamber to where the gas extraction orifice is, a small fraction of the gas that was not extracted from the gas core also exited through the liquid exit orifice and re-mixed with the liquid. This remixing created new gas-liquid mixture that had to be re-circulated to be separated again and significantly reduced the gas extraction efficiency. To address this issue, a series of off-center liquid extract orifices, which have no intersection with the gaseous core, are arranged symmetrically. Extensive experiments have shown that the gaseous core, within the operation range covered by the DYNASWIRL® phase separator, never extend to the inner boundary of the liquid exit orifices, therefore no gas collected in the gaseous will escape into the liquid exit orifices and remixing with the liquid. To further enhance the gas extraction efficiency, another gas extraction orifice, similar to the other gas extraction orifice covered with hydrophobic membrane described above, is arrange concentrically with the gaseous core surrounded by the liquid exit orifices. B. Enabling Operation with Continuous Feed Due to the remixing of the gas and liquid in the previous prototype design, it only operated in recirculation mode to ensure complete gas extraction. With the improved gas extraction design, the remixing of gas and liquid is eliminated and gas is extracted from both ends of the DYNASWIRL® chamber, with these enhancement, the operation with continuous mixture feed is implemented which is needed in the final phase separator design for tests in International Space Station (ISS). In this report period, the focus is to design a continuous feed mechanism such that the performance and operation range of the phase separator under various continuous feed conditions can be tested, and the mechanism should be easy to implement and operate in both ground based and reduced gravity environment. The continuous feed of mixture is realized by mixing air with liquid through a bubble generator, the mixture phase separator after the mixing, the void fraction of the mixture is determined by the amount of air supplied to the bubble generator. The air supply is made possible through the Venturi like bubble generator, it can create a vacuum at its throat such that air can be sucked from the atmospheric pressure (in the laboratory or in the air cabin) into the bubble generator to mix with liquid input. The bubble generator is designed to enable air suction at any flow rate that the phase separator operates at. Also the bubble generator for mixture generation is designed to create a pressure drop to absorb the pressure difference between the maximum pressure head of the pump and the maximum operation pressure of the phase separator such that accidental over-pressurization of the phase separator can be prevented. Extensive tests of the phase separator under various continuous feed conditions are under way. C. Preparation of a Test Rig for Upcoming Reduced Gravity Test A new round of reduced gravity flight tests are scheduled in the April 30th flight week, which present a great opportunity to test the improved DYNASWIRL® phase separator design and evaluate its performance. In this reporting period, we also concentrated on implementing the improved design into a test rig that is reduced gravity flight test ready. To meet the reduced gravity flight test requirement, a double containment is required to prevent any potential water leakage. All water wet parts (the phase separator, plumbing etc) are enclosed in the containment, additionally a pressure

	switch is utilized to cut off the electricity to the liquid supply pump if the pressure of the mixture supply to the phase separator is above the pre-set pressure threshold. The completed test rig has is in its final test stage and will be ready for the upcoming reduced gravity flight test. D. Instrumentation Improvement To evaluate the performance of a phase separator, it is very important to measure properly the amount of gas extracted and the void fraction of the mixture before and after the separation. An air mass flow meter is installed in the gas extraction line to measure the gas extracted from the phase separation. As in the flight test, the ABS ACOUSTIC BUBBLE SPECTROMETER® will be used to measure the low void fraction mixture after the separation from 10-3 and below. To measure high void fraction mixture input, a conductivity probe is used. In this reporting period, we continued our efforts to develop an in-house void fraction measurement device based on conductivity probes. To achieve better resolution and accuracy of small amplitude signals in noisy environment, a phase lock-in amplifier is used for the conductivity probe measurement, it is being used for ground based tests to quantify the mixture input to the phase separator under different operation conditions.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Papers from Meeting Proceedings	Wu X, Chahine GL. "Development of DYNASWIRL® Phase Separator for Space Applications." 50th AIAA Aerospace Sciences Meeting, Nashville, Tennessee, January 9-12, 2012. Proceedings of 50th AIAA Aerospace Sciences Meeting, January 2012. AIAA paper AIAA-2012-0591. , Jan-2012